Historical Consistency Predicts Mechanism Accuracy, Not Mechanism Ranking: Evidence from a Controlled Synthetic Study

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Abstract

Can historical decision consistency, measured before deployment, tell a system designer which classification mechanism to build? We test this with a pre-registered 240-condition synthetic factorial experiment (20 seeds $\times$ 6 nominal-consistency targets $\times$ 2 lexical conditions) comparing exact-match rules against fuzzy retrieval, under both a lexically clean condition and a controlled lexical/surface-form perturbation. The answer splits in two. Realized consistency predicts each mechanism's own accuracy well (Pearson $r > 0.9$ in both lexical conditions). It does not predict which mechanism wins. A frozen, prespecified consistency-threshold rule agrees with the empirically best mechanism in 100% of comparisons in one consistency band and 0% in another, because the winner is instead a constant function of the separately manipulated lexical condition, a variable the consistency signal is constructed not to observe. That the winner is retrieval, and that the exact-match rule's disadvantage *widens* rather than shrinks as consistency rises, are empirical outcomes, not consequences of how the signal is defined: the construction forces only that the signal cannot see surface-form variation, not which mechanism that variation favors, nor by how much. This finding is scoped to one synthetic generator, one lexical-perturbation model, and one motivating (non-evidentiary) production case study; it is not a deployment or generalization claim. A design-time selector built on historical consistency alone therefore cannot predict which mechanism will win in this setting. Whether a selector that also conditions on a measured representation-stability signal would reliably predict which mechanism wins is the direct next hypothesis this experiment motivates: no such selector was designed, prototyped, or tested here.

Keywords: classification-mechanism selection; historical label consistency; algorithm selection; design-time selection; representation stability.

1 Introduction

1.1 Real-World Motivation and General Problem

A production accounting-classification system applies a frozen, pre-specified decision rule to a measured historical-consistency statistic before choosing which classification mechanism to build. In one production deployment, that statistic was measured at 91.2% and the rule selected a rules-first, exact-match mechanism. In an independently generated synthetic reproduction of the same pipeline, the analogous statistic was measured at 87.56% and the same rule selected its embedding-based branch (`EMBEDDING_PRIMARY`) instead, a different mechanism from the lexical-similarity `retrieval` mechanism this paper's Experiment 1 tests (Section 3.3). No embedding-based mechanism is evaluated anywhere in this paper. Both figures sit within a few percentage points of the rule's own 90% threshold, so the two runs' decisions differ not because the underlying systems differ in kind, but because a small measurement difference happened to fall on opposite sides of a single cut-off. We report this observation as the motivation for the research question below, not as evidence for it: neither run measured whether the mechanism it selected actually performed best, and the production figures cited here are drawn from a confidential engagement, not independently reproducible from this repository.

This single observation is an instance of a general pattern: classification systems typically fix a mechanism class (an exact-match lookup, a fuzzy retrieval index, a trained model) before any measurement of the data the system will actually operate on. A statistic computed from historical decisions could, in principle, inform that choice before a system is built or deployed. Measuring data properties before choosing a system design is itself a long-studied idea, not a contribution of this paper (Section 2). What this paper studies is narrower: whether one specific such statistic, historical label consistency, actually predicts which of two qualitatively different mechanisms should be chosen, and under what conditions that prediction holds.

1.2 Research Question

The general problem of choosing among candidate procedures using measured, historical evidence has an established name: the Algorithm Selection Problem (Rice, 1976), later unified with the meta-learning literature (Smith-Miles, 2009) and, more recently, reframed by the AutoML community as automated workflow composition (Barbudo et al., 2023). This paper operates at the architecture/workflow level those literatures describe the field moving toward, not at the per-instance level Rice originally posed. Section 2 develops this positioning fully; here we note only that the problem this paper studies is a specific, narrower instance of a well-established research question, not a new one.

The question this paper asks, precisely, is: can historical decision consistency, measured before deployment, be used to select between qualitatively different classification mechanisms (exact-match rules versus retrieval-based fuzzy matching)? Four preconditions must hold in the target setting. There must be a history of repeated decisions on the same or similar items, observable ground-truth labels for those decisions, a consistency statistic that can be measured from them (formalized in Section 3.2 as the Automated Determinism Score, ADS), and sufficient historical coverage for that statistic to be estimated reliably. This paper's experiment is constructed to satisfy all four by design; where a real system does not, the question this paper answers does not directly apply to it.

The intuition motivating this question is straightforward. If a product has almost always been booked to the same account historically, that looks like strong evidence the mapping is already solved, and a simple, precedent-based mechanism, an exact lookup against history, should suffice; conversely, if the historical mapping is inconsistent, a more flexible mechanism should be needed. The natural hypothesis this suggests is that higher historical consistency should favor a simpler, precedent-based mechanism over a more flexible one. This is exactly the hypothesis this paper tests, and, as Section 5 reports, partly falsifies.

1.3 Experimental Approach and Main Findings

We test this with a pre-registered, 240-condition synthetic factorial experiment. Two mechanisms are compared in isolation: exact-match rules and rapidfuzz-based fuzzy retrieval (cutoff calibrated once, before the run, to 75). Two lexical conditions, CLEAN (no surface-form perturbation) and VARIED (a fixed, controlled lexical/surface-form perturbation), are crossed

with six nominal historical-consistency targets and twenty random seeds. A third mechanism, a large language model, is deliberately excluded from the primary comparison. The reason is documented in Section 7.3, and it is not convenience. The synthetic generator's product string carries no signal predictive of the true label, which would make any LLM result uninterpretable as evidence about historical consistency.

The experiment produces a two-part finding that this paper keeps explicitly separate throughout. First, realized historical consistency strongly predicts each mechanism's own accuracy: both the rules and the retrieval mechanism become more accurate as consistency rises, in both lexical conditions. Second, realized historical consistency does not predict *which* mechanism wins: the mechanism that actually performs best in a given condition is instead a constant function of the (separately manipulated) lexical condition, unrelated to how consistent the underlying data is. These are two different claims with two different answers. Here the answers are opposite. This paper never merges them into a single statement about whether "historical consistency works."

The second finding has a feature worth isolating early, because it is the part a reader might otherwise assume the experiment's construction guarantees. That realized consistency cannot *see* the lexical condition is indeed true by construction (Section 4.3). But which mechanism the lexical condition favors, and how that advantage scales with consistency, are not fixed in advance by any definition. The empirical answer is counterintuitive: under perturbation, retrieval wins in every condition, and the exact-match rule's disadvantage *widens* as realized consistency rises (-0.137 to -0.185 whole-set-accuracy points across ADS bands; Section 5.5), rather than shrinking as the precedent-based intuition would predict. High-consistency data is exactly where an exact-match rule appears safest and is in fact where it has the most to lose when surface strings are perturbed. That direction and magnitude are the empirical core of this paper; the accuracy-versus-ranking distinction organizes them.

1.4 Contribution Statement and Paper Roadmap

This paper's contribution is a precise, evidenced boundary on what a historical-consistency signal can and cannot tell a system designer, not a new metric, a new architecture, or a validated selection method. Formally: in this synthetic product-classification generator, realized historical decision consistency is strongly predictive of each classification mechanism's own accuracy (exact-match rules and fuzzy retrieval alike; Pearson $r \approx 0.91$–0.96 in both

lexical conditions). It is *not*, however, predictive of which of the two mechanisms will outperform the other: under a controlled lexical perturbation, retrieval wins on whole-set accuracy in every tested noisy condition, and the two mechanisms are statistically indistinguishable in every tested noise-free condition, in both cases independent of realized consistency across its full observed range. A pre-specified, consistency-only decision rule consequently agrees with the empirical winner only in the sub-range where its own threshold-driven recommendation happens to coincide with the constant true winner, because the consistency signal is computed from a stable product identity that cannot observe the surface-form instability that actually determines the outcome here. This is not a new metric (Section 2), not a new architecture, and not a method we claim selects the right mechanism. It is a bounded empirical finding about where one specific, already-used signal succeeds and where it fails.

1.5 Scope and Non-Novelty

Four things this paper does not claim, stated once here rather than repeated at each point of contact. ADS is not a new metric: its closed form is cluster purity, and equally the raw majority-vote agreement proportion (Section 2.1). Design-time selection from historical evidence is not a new pattern; it is the Algorithm Selection Problem (Rice, 1976) and its descendants. R3 is not a proposed method: it is reused unmodified from a production system's own rule taxonomy, and this paper measures where it fails. The application domain is not unprecedented; at least one non-peer-reviewed industry source (Ken From Finance, 2025) describes similar practice already in use. What the paper does claim is a bounded empirical result about where one already-used signal succeeds and where it fails.

1.6 Paper Roadmap

Section 2 positions this work against the algorithm-selection, meta-learning, AutoML, self-designed-systems, reject-option, learning-to-defer, and model-cascade literatures, with Table 1 setting out the prior art and the narrow delta row by row. Section 3 defines the historical-consistency signal and the accuracy/ranking distinction the rest of the paper is organized around. Section 4 describes the experimental design in full. Section 5 reports results, keeping the accuracy and ranking findings in separate subsections. Section 6 discusses what the finding does and does not license, including the production case study's proper (non-evidentiary) role. Section 7 states the

experiment's limitations; Section 8 names, without building or testing, the natural next research direction; Section 9 concludes.

2 Related Work

2.1 Cluster Purity and Majority-Vote Agreement

The historical-consistency signal this paper studies, the Automated Determinism Score (ADS, Section 3), is mathematically identical to an existing, well-established quantity. Cluster purity (Manning et al., 2008; Amigó et al., 2009) is defined as $\text{purity}(\text{cluster}_k) = \max_j(\text{count of class } j \text{ in cluster } k)/|\text{cluster}_k|$, the identical closed-form expression as ADS under the substitution "cluster" $\leftrightarrow$ "an item's historical booking multiset." Independently, the raw majority-vote agreement proportion used throughout the crowdsourcing and truth-inference literature descending from Dawid and Skene (1979) is the same quantity under a different name, applied to the same kind of repeated-labeling data. The paper's contribution lies in what this already-known metric is used for, not in its formula.

2.2 Design-Time Algorithm and Architecture Selection

The Algorithm Selection Problem (Rice, 1976) originates the general question of choosing a procedure from historical evidence, selecting a single algorithm per problem instance. Smith-Miles (2009) unifies this framing with meta-learning, and later work applies the same design-time, historical-evidence-driven pattern to select a generic machine-learning algorithm within the same broad application area, classification, that this paper studies. This general pattern is well established. This paper's narrower question is whether a *label-consistency-specific* signal, as opposed to generic statistical meta-features or benchmarked performance search, predicts a *qualitative mechanism-class* choice (rules versus retrieval, not one machine-learning algorithm versus another), and it reports precisely where that narrower claim holds and where it fails.

Barbudo et al. (2023) documents the field's shift from single-algorithm selection toward automated, benchmarked-performance-driven workflow composition, describing the field as having moved from the algorithm selection problem toward the broader challenge of composing an entire processing workflow, the closest terminological echo of this paper's own framing found in the surrounding literature. The delta is one of evidence type and mechanism, not performance: this paper studies a label-consistency signal driv-

ing a named, interpretable threshold rule, not a benchmarked optimization search over a combinatorial pipeline space, a difference in kind rather than a claim of superiority.

Idreos and Kraska (2019) describes the strongest non-machine-learning structural analog to this paper's design-time framing: measure historical evidence once, choose system composition before serving traffic. The evidence type there is workload or query *frequency*, not label *consistency*, and the literature's own thesis is continuous self-adaptation, explicitly rejecting a one-time, human-gated design decision, the opposite temporal structure from the one this paper's experiment tests. This literature argues against the one-shot gate; it is cited here as contrast, not as an unsolved problem this paper addresses. This paper's experiment is about a specific signal type (label consistency) and its specific limitation (blindness to representation instability), not a defense of the one-shot-gate design pattern in general.

2.3 Inference-Time Selection and Escalation

Chow (1970) originates the single-threshold accept/reject decision on live posterior confidence, formalized further by El-Yaniv and Wiener (2010)'s risk-coverage framework and surveyed comprehensively by Hendrickx et al. (2024). This literature is not directly engaged by Experiment 1, because Experiment 1 tests a *design-time* selection rule applied once before deployment, not a *runtime* reject threshold applied per item at inference time; we cite it only to establish that distinction, which this paper's scope sits firmly on the design-time side of.

Mozannar and Sontag (2020) establishes the modern learning-to-defer framework: a trained or rule-based function decides, per item, whether to defer to a human, informed by historical decision data. As with reject-option methods, this literature establishes the same design-time/inference-time boundary this paper's scope sits on the design-time side of; the production system's runtime human-escalation cascade sits closer to this literature, but that cascade is not evaluated in Experiment 1, which studies only the isolated rules and retrieval mechanisms.

FrugalGPT (Chen et al., 2024) is the closest structural match in the surrounding literature to a multi-tier, historically-calibrated cascade of classification mechanisms. RankGPT (Sun et al., 2023) establishes that using a large language model to re-rank a pre-fetched candidate list, rather than to classify from a blank input, is commodity information-retrieval technique. Neither family is directly relevant to the experiment reported here (the LLM mechanism is explicitly excluded from Experiment 1, Section 7.3), and both

are cited only for completeness of the design-time/runtime distinction this paper's framing depends on.

2.4 Domain-Specific Practice

Jørgensen and Igel (2021) empirically demonstrates, in the same application domain (financial transaction-to-account classification), that a global classifier generalizes far worse across companies than per-company models, the same phenomenon the production system's own cross-company-consistency threshold encodes, though reported there as a direct accuracy gap rather than formalized into a named metric. Independent industry sources (accounts-payable automation vendors, not peer-reviewed) (Ken From Finance, 2025; Peakflo, 2025; Ramp, 2025) describe similar informal historical-consistency-audit practice already in production use. The narrow delta that survives, stated row by row in Table 1, is academic rather than methodological. No peer-reviewed study of this specific document-classification niche was found in the literature search this paper's evidence base draws on. That is a modest niche-novelty observation, much weaker than any claim of methodological novelty, and not this paper's scientific contribution.

3 Problem Setting and Signal Definition

3.1 Historical Product-Level Decisions and Account Labels

The data object every mechanism and every signal in this paper operates on is a repeated, historically observed mapping from an item (a "product") to a label (an "account"), together with an occurrence count. This formulation is domain-agnostic: nothing in the definitions below references Romania, fiscal documents, or invoices specifically, even though the motivating case study (Section 1.1) and the synthetic generator (Section 4) both instantiate it in that domain. Studying this question in a given setting requires the four preconditions named in Section 1.2 to hold: a history of repeated decisions, observable labels for those decisions, a consistency statistic that can be measured from them, and sufficient historical coverage for that statistic to be estimated reliably.

3.2 The Automated Determinism Score (ADS)

For an item p with historical occurrence counts $c_1, \ldots, c_k$ across its k observed labels, the Automated Determinism Score is

Table 1: Related-work positioning. Rows are the research families closest to this paper's design-time consistency-driven mechanism-selection question; the narrow delta column states only what Section 5's experiment actually bears on, never a claim about the cited literature being incomplete.

Research family	Prior art	Narrow delta
Cluster purity / majority-vote agreement	Defines the identical closed-form expression ADS uses	None at the metric level; delta is entirely in what the metric is used for
Algorithm Selection / meta-learning	Selects a single algorithm per instance or dataset from historical/meta-feature evidence	Studies a label-consistency-specific signal driving a qualitative mechanism-class choice, not a generic-algorithm choice
AutoML / workflow composition	Automated, benchmarked-performance-driven search over pipeline configurations	Interpretable threshold rule vs. optimization search; evidence type differs, not claimed superior
Self-designed / instance-optimized systems	Continuous, workload-frequency-driven self-adaptation, explicitly rejecting a one-shot gate	Studies a one-shot, label-consistency signal and its specific representation-stability blind spot
Reject option / selective classification	Single runtime accept/reject threshold on live confidence	Not engaged directly; establishes the design-time/inference-time boundary this paper sits on one side of
Learning to defer	Trained/rule-based per-item deferral informed by historical decisions	Same boundary as above; the production cascade (not tested here) sits closer to this literature
Model cascades / LLM routing	Historically-calibrated multi-tier escalation across models	Not engaged; LLM excluded from Experiment 1 by design (Sec. 7.3)
Domain-specific practice	Empirically documents the same cross-company phenomenon and similar informal audit practice in the same domain	Domain remains academically under-served; not a methodological novelty claim

$$\mathrm{ADS}(p) \;=\; \frac{\max_i c_i}{\sum_i c_i} \tag{1}$$

the empirical probability that a random historical occurrence of p was assigned its modal label. This is mathematically identical to cluster purity (Manning et al., 2008; Amigó et al., 2009) and to the raw majority-vote-agreement baseline used throughout the crowdsourcing literature descending from Dawid and Skene (1979) (Section 2.1); it is a descriptive statistic, not a novel contribution of this paper. Two aggregates of the per-item score in Equation (1) over a product population are used later in this paper only as diagnostic covariates, never as the primary independent variable: the occurrence-weighted mean (what a random transaction actually experiences) and the unweighted mean (the catalog's intrinsic learnability). The primary independent variable throughout Sections 4 and 5 is a third quantity, *realized* ADS, introduced precisely in Section 4.3.

3.3 Two Compared Mechanisms: Exact-Match Rules and Fuzzy Retrieval

The first of the two mechanisms this paper compares, `rules`, is an exact company-scoped string lookup against the historical training data, falling back to an exact global lookup if no company-scoped match exists; any hit is returned as the mechanism's answer, and no hit results in abstention. There is no confidence threshold to calibrate for this mechanism: matching is binary, exact-match or no-match. Under the whole-set-accuracy metric used throughout this paper (Section 4.7), an abstention counts as incorrect, with no partial credit. Coverage, the share of test items for which the mechanism does not abstain, is below 1.0 in every one of the 240 conditions reported in Section 5, a real, reported asymmetry with the retrieval mechanism below, not one this paper normalizes away. This isolated mechanism is a simplified test-harness component built for Experiment 1; it is not the shipped production cascade's first tier, which additionally applies confidence gating and human-review escalation logic outside this experiment's scope.

The second mechanism, `retrieval`, applies a rapidfuzz-based lexical-similarity match (the `WRatio` scorer), company-scoped then global, as the *primary* mechanism for every test item, rather than as a fallback gated behind a rules-miss. A match is accepted if its similarity score meets a cutoff calibrated once, before the frozen run, on a dedicated calibration dataset disjoint from every condition reported in Section 5 (Section 4.5); items below the cutoff abstain. Coverage is 1.0 in every one of the 240 conditions: this

mechanism never abstains at the calibrated cutoff. We deliberately call this mechanism "retrieval," not "embedding," because it is a lexical-similarity match over surface strings, not a semantic embedding model: no embedding model was trained, downloaded, or evaluated anywhere in this experiment, and describing it as "embedding-primary" would misstate what was actually tested.

3.4 Mechanism Accuracy vs. Mechanism Ranking

This paper is organized around a distinction that Section 5 keeps in separate subsections throughout: "ADS predicts mechanism accuracy" and "ADS predicts mechanism ranking" are different claims, with different (here, opposite) truth values, and this paper never merges them. The first is a correlational claim about a single mechanism's own performance level as realized consistency varies (Section 5.2); the second is a claim about *which* of the two mechanisms wins a head-to-head comparison (Section 5.3). A signal can be strongly informative about the first while carrying no information about the second, precisely if the quantity that determines relative ranking is one the signal cannot observe, which is exactly what Section 5 reports.

4 Experimental Design

4.1 Research Hypothesis

H1 (revised) was stated and pre-registered before any data from the frozen run existed. Under controlled synthetic conditions, higher realized historical decision consistency will be associated with predictable changes in relative mechanism performance, such that a pre-specified, frozen consistency-based decision rule (R3, defined in Section 4.7) will agree with the empirically best-performing mechanism more often than chance. This phrasing is deliberately weaker than an earlier draft that required monotonic dominance in every consistency band; H1 claims only better-than-chance agreement, with falsification criteria fixed in advance (Section 4.8).

4.2 Synthetic Generator and Factorial Design

Every condition is generated from a synthetic invoice-classification generator spanning 60 companies and a 1,200-product vocabulary, fixed across every condition. Each generation call draws its own independent, reproducible random stream (`random.Random(seed)`) rather than mutating one

shared global generator, so that a 20-seed sweep produces statistically independent replicates. Cross-company alignment, the rate at which different companies agree on a shared product's dominant account, is fixed at 0.695 (the production-observed value, cited from a confidential engagement and not independently reproducible from this repository), so that it does not become a second, uncontrolled independent variable alongside the consistency sweep itself. This generator is not calibrated against real invoice or OCR error data; Section 7 states this limitation explicitly.

Two factors are varied systematically. The first is the generator's nominal target for the share of products intended to be near-deterministic, run at six levels: $\{0.00, 0.20, 0.30, 0.50, 0.75, 1.00\}$. The second is the lexical condition, CLEAN or VARIED, two levels. Crossed with 20 seeds, this produces $6 \times 2 \times 20 = 240$ conditions exactly (Figure 1), the frozen final run this paper reports. A separate, wider 17-point grid was used only during an earlier, mechanism-blind calibration pass (Section 4.8) to characterize the generator's realizable range; it is not itself part of the 240-condition frozen run and its data does not appear in Section 5.

4.3 Realized ADS vs. Target Deterministic Share

The generator's nominal target is an input knob at generation time, not itself the scientific independent variable. The independent variable this paper actually reports against, throughout Section 5, is *realized* ADS: the measured, train-only share of products whose per-product determinism score exceeds 0.95, computed after generation, grouped by the stable product identity. Realized ADS is the exact quantity the frozen decision rule R3 (Section 4.7) thresholds; the two usually, but do not always, coincide with the nominal target. At the target of 0.50, for instance, individual seeds' realized ADS straddles R3's 0.70 boundary (observed range 0.659–0.727 across 20 seeds), so seeds sharing one nominal target can land in different R3 bands. Across the frozen run, realized ADS ranges from 0.44 to 0.93, structurally capped by the fixed cross-company-alignment parameter (Section 4.2): the region of realized ADS at or above roughly 0.93 was never reachable by this generator and is therefore untested, not merely unfavorable. Realized ADS is invariant to the lexical condition by construction, since it is computed by grouping on the stable product identity, never on the perturbable surface string. This is confirmed directly in the data (byte-identical realized-ADS means and standard deviations between the CLEAN and VARIED conditions at every target). Every result, table, and figure in Section 5 uses realized ADS, never the nominal target, as its independent-variable axis.

4.4 Lexical Conditions: CLEAN and VARIED

Under CLEAN, the generator's product surface strings are emitted without any lexical perturbation, the existing generator's unmodified default output. Under this condition, the rules and retrieval mechanisms are statistically indistinguishable (within a pre-registered practical-equivalence margin of $\delta = 0.02$ whole-set-accuracy points) at every one of the six targets: zero of the 120 CLEAN conditions produce a defined empirical winner (Section 5.3). This is evidence that the two mechanisms perform almost identically specifically when there is no lexical noise for retrieval's fuzzy matching to be robust to, for this generator, rather than a general claim that the two mechanisms are equivalent.

Under VARIED, a fixed, five-transform perturbation algorithm is applied, independently per invoice line, only to the display/lookup string; the ground-truth product identity used to build the mechanisms' knowledge bases and to score accuracy is untouched, so semantic identity is preserved by construction, not by convention. The five transform types are case variation, punctuation variation, token reorder, abbreviation, and whitespace variation. Each line is perturbed with probability $P_{\text{TRANSFORM}} = 0.3$, a value chosen once, before the frozen run, by a pilot sweep against a target corruption-share acceptance band, not derived from measured real-world OCR or typo error rates and never re-tuned after seeing mechanism-accuracy results. The perturbation is fully reproducible: a deterministic per-line subseed derivation reproduces the identical transform decision on any re-run of the same seed.

4.5 Mechanisms and Retrieval Cutoff

Only the retrieval mechanism has a tunable parameter, its similarity cutoff; the rules mechanism has none. The cutoff was calibrated once, on a dedicated generation run with its own seed, distinct from every seed used in the 240-condition frozen sweep, before that sweep was run. Candidate cutoffs $\{70, 75, 80, 85, 90\}$ were evaluated against a product-identity hit-rate criterion, subject to a minimum coverage floor of 30%. That criterion asks whether the top fuzzy match resolves to the same underlying product as the query, not whether the resulting account label is correct. A cutoff of 75 was selected and frozen for the entire run, never re-tuned per band, per lexical condition, or after seeing accuracy results.

4.6 Seed Structure, Train/Test Separation, and Winner Definition

Twenty seeds, $\{31001, \ldots, 31020\}$, are used identically across every one of the twelve (target, lexical) cells, for 240 total conditions. The seed range is a documented, deterministic rule, fixed before the run and never hand-picked after seeing any result.

Realized ADS is computed exclusively from each condition's train split, mirroring real deployment temporal logic: a design-time decision can only see history, never the held-out future it will later be judged against. Each mechanism's knowledge base is likewise built only from that condition's train rows. Every accuracy and coverage number reported in Section 5 is computed exclusively on that condition's test rows. This separation is verified in code, not merely asserted: the consistency computation filters to train-split rows before any aggregation, and no test-split label ever reaches the mechanism-selection rule.

Within a condition, both mechanisms are scored on identical held-out test lines: a paired rather than an independent comparison. The empirical winner, stated formally in Equation (2), is defined by a paired bootstrap on the accuracy difference $\text{acc}_{\text{rules}} - \text{acc}_{\text{retrieval}}$ (2,000 resamples, 95% percentile interval): rules wins if the entire confidence interval lies above $+\delta$; retrieval wins if it lies entirely below $-\delta$; otherwise the condition is scored a tie, a distinct category never broken toward either mechanism. The practical-equivalence margin $\delta = 0.02$ was fixed at this calibration stage, before the frozen run, and is not re-examined against the outcome data anywhere in this paper.

$$\text{winner} = \begin{cases} \text{rules} & \text{if } \text{CI}_{95\%}(\text{acc}_{\text{rules}} - \text{acc}_{\text{retrieval}}) \text{ entirely} > +\delta \\ \text{retrieval} & \text{if } \text{CI}_{95\%}(\text{acc}_{\text{rules}} - \text{acc}_{\text{retrieval}}) \text{ entirely} < -\delta \\ \text{tie} & \text{otherwise} \end{cases} \tag{2}$$

4.7 Statistical Analysis

The primary performance metric for each mechanism is whole-set accuracy: a prediction counts as correct only if the mechanism both answered and was right, with abstention counted as incorrect and no partial credit. The agreement rate between the frozen decision rule R3 (below) and the empirical winner is the number of conditions where R3's selected mechanism equals the empirical winner, divided by the number of conditions with a *defined*

comparison, excluding ties and conditions where R3 selects the excluded LLM band. Because the LLM mechanism is excluded from the primary comparison, R3 discriminates between exactly two mechanisms wherever a comparison is defined, so the chance baseline used throughout this paper is 0.5, not one-third. Agreement rates are tested against this chance baseline with an exact two-sided binomial test, and reported with a Wilson 95% confidence interval.

The frozen decision rule R3, so named in the production system's own internal rule taxonomy rather than an alternative considered within this experiment, is reused unmodified from that system's architecture-decision procedure and is not re-derived or re-tuned for this experiment. It maps realized ADS to a mechanism recommendation by a fixed threshold: at or above 0.90, rules; between 0.70 and 0.90, retrieval; below 0.70, the excluded LLM band.

4.8 Calibration and Preregistration

Three calibration gates were closed before the frozen run. First, a mechanism-blind ADS calibration swept a dense 17-point target grid with ten seeds per target, using no import of and no reference to either mechanism's code. It characterized the generator's realizable realized-ADS range and chose the six final targets by a greedy furthest-point selection over that curve, so that no target band was chosen with foreknowledge of mechanism behavior. Second, the retrieval cutoff was calibrated as described in Section 4.5. Third, the empirical-winner, tie, and confidence-interval definitions (Section 4.6) were fixed. The 17-point calibration grid is wider than, and distinct from, the six-target grid actually used in the 240-condition frozen run (Section 4.2); its own data does not appear in Section 5.

The falsification table used to classify the experiment's outcome (H1 SUPPORTED / PARTIALLY SUPPORTED / NOT SUPPORTED / INCONCLUSIVE) was fixed in advance of the frozen run. The observed pattern matches the pre-registered PARTIALLY SUPPORTED row exactly (Section 6.3), which is not a post-hoc reinterpretation of the data but one of four possible outcomes specified before any data existed. No threshold, mechanism definition, retrieval cutoff, winner definition, or generator parameter was changed after the frozen run's results were known; the post-hoc analysis this paper's Results section also draws on explains the frozen data, it does not regenerate or re-tune it.

4.9 Reproducibility

The entire generator, both mechanisms, the lexical perturbation model, and the calibration and analysis code are public, offline, require no API keys or external cost, and are covered by an automated test suite. The 240-condition dataset itself is generated, not collected: re-running the committed seed manifest under the frozen configuration of Table 2 via `python scripts/experiments/exp1/run_final.py` reproduces the results reported in Section 5 exactly. By contrast, the production case study discussed in Sections 1.1 and 6.2 is confidential: only cited aggregate statistics appear in this paper, never row-level data, and it is not reproducible from this repository in any sense. A reader can therefore reproduce this paper's entire statistical evidence base for the accuracy and ranking findings end-to-end from public code with no client data, but cannot reproduce the motivating production observation, which is cited, not offered as something to re-run.

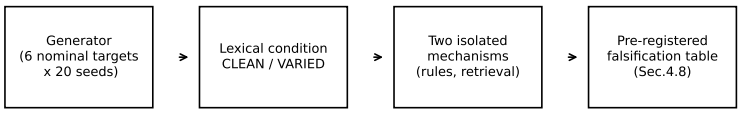


Figure 1: Experimental design and pre-registration flow. Shows the factorial structure and the pre-registration-to-falsification-table pipeline before any result is shown, so a reader can assess whether the eventual result could have come out differently.

5 Results

5.1 Experimental Completeness and Frozen Design

All 240 conditions succeeded; none failed. All 240 (seed, target, lexical) triples are unique, with no duplicates. Every frozen-configuration assertion built into the run script passed: $\delta = 0.02$, retrieval cutoff = 75,

Table 2: Experimental configuration, frozen before the 240-condition run.

Parameter	Value
Nominal targets	0.00, 0.20, 0.30, 0.50, 0.75, 1.00
Lexical conditions	CLEAN (p=0.0), VARIED (p=0.3)
Seeds	20 (31001–31020), identical across cells
Retrieval cutoff	75
R3 thresholds	rules ≥ 0.90; retrieval $[0.70, 0.90)$; excluded < 0.70
Winner rule	paired bootstrap, δ=0.02, 2,000 resamples, α=0.05
Primary metric	whole-set accuracy (abstention = incorrect)
LLM mechanism	excluded from primary comparison
Total conditions	$6 \times 2 \times 20 = 240$

$P_{\text{TRANSFORM}} = 0.3$, 20 seeds, and R3 thresholds 0.90/0.70. The full harness test suite passed on the code version that produced this data, and that code is unmodified (byte-identical to the frozen commit) as of this draft.

5.2 ADS Predicts Individual Mechanism Accuracy

Realized ADS is strongly correlated with each mechanism's own accuracy, in both lexical conditions and reported for both mechanisms rather than only the stronger one (Figure 2): Pearson $r(\text{realized ADS}, \text{rules accuracy}) \approx 0.909$ (VARIED) and 0.959 (CLEAN); $r(\text{realized ADS}, \text{retrieval accuracy}) \approx 0.948$ (VARIED) and 0.955 (CLEAN). This is a correlational finding about each mechanism's own performance level. It says nothing yet about which mechanism performs better than the other; that is the subject of Section 5.3.

5.3 ADS Does Not Predict Mechanism Ranking

Realized ADS does not predict which mechanism wins. Under VARIED, the empirical winner is retrieval in all 120 of 120 conditions, exceptionless and unconditional on realized ADS across its full observed range (0.44–0.93). Under CLEAN, the empirical winner is a tie in all 120 of 120 conditions, also exceptionless (Figure 3). The lexical condition here is a manipulated, controlled experimental factor, not merely an observational correlate, which is what licenses describing the empirical winner as *governed by* the lexical condition in Section 6, rather than only correlated with it. Because realized ADS is invariant to the lexical condition by construction (Section 4.3), and the empirical winner is a constant function of the lexical condition, ADS

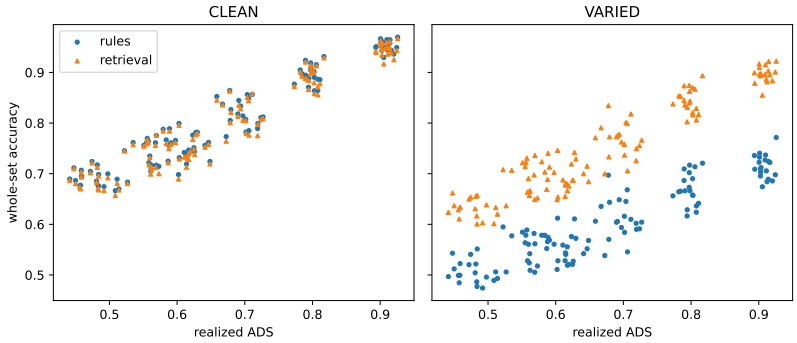


Figure 2: Realized ADS versus each mechanism's own whole-set accuracy, visualizing the Section 5.2 correlations directly. Source: data/outputs/experiments/exp1/final/final_condition_results.csv.

structurally cannot track the quantity that actually determines the winner in this experiment.

5.4 R3 Threshold Agreement by Realized-ADS Region

This subsection reports the paper's headline finding, leading with the band structure rather than the flat aggregate. The primary framing bins by each row's own realized ADS against R3's own thresholds, matching the locked wording this paper draws on. Under it, R3 agrees with the empirical winner in 100% (32 of 32) of conditions in the realized 0.70–0.90 band, and in 0% (0 of 18) of conditions in the realized ≥ 0.90 band (Figure 4). A secondary, also valid framing bins by nominal target rather than each row's own realized value: 30 of 30 (100%) at nominal targets 0.50 and 0.75, and 2 of 20 (10%) at nominal target 1.00. These two framings differ because nominal target and per-row realized ADS do not always coincide (Section 4.3): at nominal target 1.00, two of the twenty seeds realize just below the 0.90 boundary and therefore fall in the 0.70–0.90 band under the per-row framing while sharing the 1.00 nominal target under the by-target framing. We report both, explicitly labeled, rather than silently switching between them.

Only after this band structure do we report the flat aggregate: 32 of 50 defined comparisons agree (64.0%), Wilson 95% confidence interval $[50.14\%, 75.86\%]$, exact binomial $p = 0.0649$ against a 50% chance baseline, which is not significant at $\alpha = 0.05$ (Table 3). This aggregate is a near-cancellation of two individually far more decisive, opposite patterns: the

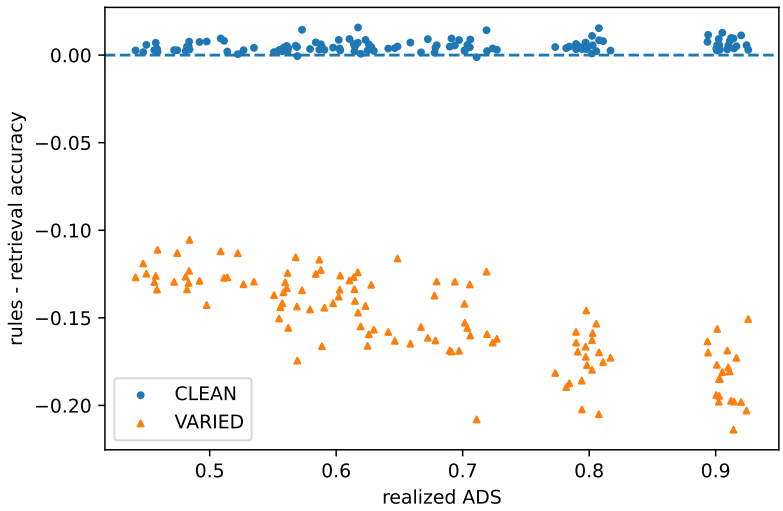


Figure 3: Mechanism-ranking constancy across the lexical condition: the rules-minus-retrieval accuracy difference never crosses zero within a lexical condition, regardless of realized ADS. Source: data/outputs/experiments/exp1/final/final_condition_results.csv.

per-row realized-ADS bands are exceptionless (32 of 32 and 0 of 18), and the corresponding by-nominal-target counts are independently significant at exact binomial $p = 1.9 \times 10^{-9}$ (30 of 30) and $p = 4.0 \times 10^{-4}$ (2 of 20) respectively (Section 5.6). This is not a single moderate effect, and reporting the aggregate alone would obscure exactly the structure this subsection leads with. The CLEAN condition contributes zero defined comparisons at any band: every measurable piece of evidence about R3's agreement with the empirical winner comes from the VARIED condition.

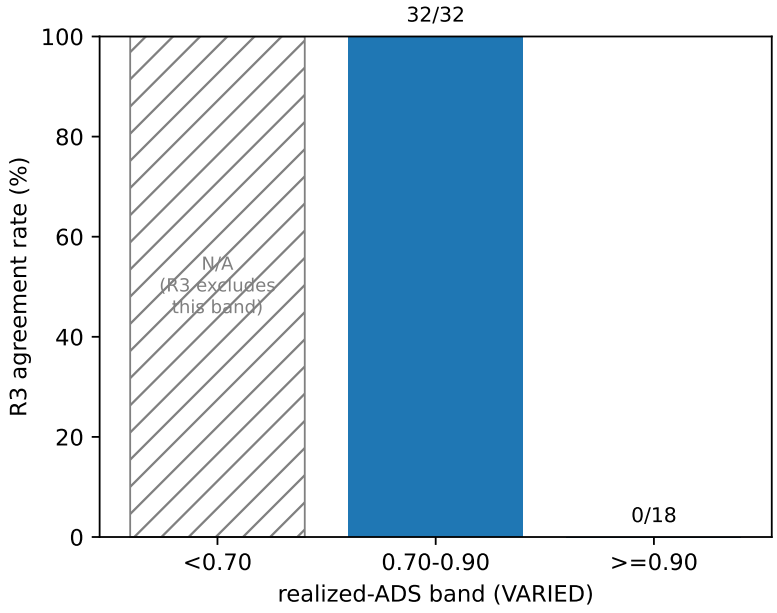


Figure 4: R3 agreement with the empirical winner by realized-ADS region: 100% (32/32) in the 0.70–0.90 band versus 0% (0/18) in the ≥ 0.90 band.

5.5 ADS × Representation-Stability Interaction

Table 4 shows the mechanistic pattern behind Section 5.4's reversal. Under VARIED, the rules-minus-retrieval accuracy gap widens as realized ADS rises, from -0.137 in the < 0.70 band to -0.168 in the 0.70–0.90 band to -0.185 in the ≥ 0.90 band, the opposite of what R3's threshold design assumes. Under CLEAN, the gap stays small and positive, $+0.005$ to $+0.006$, across all three bands: near-equivalence throughout, absent lexical noise.

Table 3: Main results: overall agreement, the by-nominal-target realized-ADS split (the framing with an independently reported binomial test in the frozen evidence), and the Section 5.2 correlation ranges. The sharper, exceptionless per-row realized-ADS-band counts (32/32, 0/18) are reported in Section 5.4; no independently frozen p-value exists for those exact counts, so none is stated here.

Quantity	Value	95% CI	p (vs. 50%)
Overall agreement (32/50)	64.0%	[50.14%, 75.86%]	0.0649
By target, targets 0.50/0.75 (30/30)	100%	—	1.9×10^{-9}
By target, target 1.00 (2/20)	10%	—	4.0×10^{-4}
Pearson r(realized ADS, rules accuracy): 0.909 (VARIED) / 0.959 (CLEAN)			
Pearson r(realized ADS, retrieval accuracy): 0.948 (VARIED) / 0.955 (CLEAN)			

This table is the direct mechanical explanation for why R3's agreement rate is 100% in the middle band and 0% at the top: R3 succeeds only where its own ADS-driven recommendation happens to equal the constant true winner. We report this pattern here; the causal account for why it occurs is developed in Section 6.4, not here, since that account rests on additional, explicitly post-hoc reasoning this subsection does not need.

Table 4: Mechanism-winner behavior by realized-ADS region, both lexical conditions. Mean whole-set accuracy and the rules-minus-retrieval gap.

Realized ADS band	Lexical	n	Rules acc.	Retrieval acc.	Gap
< 0.70	CLEAN	70	0.7431	0.7383	+0.0048
< 0.70	VARIED	70	0.5493	0.6867	−0.1374
0.70–0.90	CLEAN	32	0.8742	0.8687	+0.0055
0.70–0.90	VARIED	32	0.6529	0.8207	−0.1678
≥ 0.90	CLEAN	18	0.9534	0.9471	+0.0063
≥ 0.90	VARIED	18	0.7130	0.8980	−0.1851

5.6 Statistical Interpretation

Both individual by-nominal-target bands are far more statistically significant than the flat aggregate ($p = 1.9 \times 10^{-9}$ for the 30/30 retrieval-region band and $p = 4.0 \times 10^{-4}$ for the 2/20 rules-region band, versus the aggregate's $p = 0.0649$). The sharper per-row realized-ADS bands (32/32 and 0/18,

Section 5.4) are exceptionless but, as noted there, are not independently paired with a frozen p-value anywhere in this paper's evidence base. The number of defined comparisons is small in both informative bands (30 and 20 by the by-target framing; 32 and 18 by the per-row framing), enough for the binomial tests above to be decisive but not enough to subdivide further. The practical-equivalence margin $\delta = 0.02$, frozen before the run, is the specific reason the CLEAN condition contributes zero defined comparisons; we do not relax it or reinterpret the frozen data to reach a different verdict anywhere in this paper.

5.7 Summary of Findings

Realized ADS strongly predicts each mechanism's own accuracy (Section 5.2); it does not predict which mechanism wins (Section 5.3); the mechanistic reason is the ADS-by-representation-stability interaction reported in Section 5.5; and the flat 64.0% aggregate agreement rate alone under-communicates the underlying 100%-versus-0% band structure (Section 5.4). These four statements, taken together, are what Section 6 discusses; none of them is new information beyond what the preceding subsections already established.

6 Discussion

6.1 What the Experiment Supports

In this synthetic experiment, historical decision consistency is informative about classification-mechanism *difficulty*, not mechanism *ranking*, when, as observed here, ranking is governed by a representation-stability property the consistency signal does not observe. This is a narrower, more specific, empirically falsifiable (and partly falsified) refinement of the original hypothesis that historical consistency alone selects the right mechanism, not a confirmation of that hypothesis.

6.2 The Production Case Study in Light of These Results

The production observation described in Section 1.1 is consistent with Section 5.2's positive finding. A real deployment did show a measured consistency statistic correlating with something real about its own data. It was never itself a controlled test of Section 5.3's ranking finding, though, because production never ran a lexical-noise sweep of the kind this experiment's VARIED condition applies. This experiment does not confirm the

production observation; it investigates the more general question that observation raised. The production consistency figures cited in Section 1.1 remain, as stated there, generated by an earlier version of the measurement pipeline and are cited from a confidential engagement, not independently reproducible from this repository, a caveat repeated here rather than assumed to carry forward for a reader who begins at this section.

6.3 What the Original Hypothesis Got Right and Wrong

H1's deliberately weaker form (better-than-chance agreement rather than monotonic dominance in every band) is exactly why a PARTIALLY_SUPPORTED verdict was a coherent, pre-registered possible outcome rather than a post-hoc excuse. The point estimate, 64.0%, is directionally above chance in the only slice where agreement is measurable at all (VARIED), and one full realized-ADS band shows perfect, exceptionless agreement (32 of 32, Section 5.4), with the corresponding by-nominal-target count independently significant at $p = 1.9 \times 10^{-9}$ (Section 5.6). This result is not without value, but it is not evidence that H1 as originally intended was confirmed.

The unconditional form of the original hypothesis, that ADS thresholds alone select the right mechanism, is falsified, exceptionlessly, in the realized ≥ 0.90 band. This is not a minor or marginal failure: it is the exact band in which the frozen decision rule is most confident, since higher realized consistency produces R3's strongest recommendation for rules, and it is precisely there that the rule disagrees with the empirical winner in every single one of 18 tested conditions.

6.4 Mechanistic Explanation: Representation Stability

The evidence is consistent with a specific mechanistic account. Realized ADS is computed on the stable product identity, structurally blind to the perturbable surface string; in this experiment, which mechanism wins is a constant function of the lexical condition, a variable ADS cannot observe by construction. We state this account explicitly as inferred from exhaustive but post-hoc inspection of the frozen data and the generator's code. It is not the result of a second, independently designed confirmatory experiment. The pattern holds without exception across all 240 rows, but a fresh, prospective test of this exact causal claim on new data would strengthen it further (Section 7.1).

Table 4's widening gap reveals a specific asymmetry: retrieval's accuracy cost under lexical noise stays comparatively flat across realized-ADS bands,

while rules' cost grows with realized ADS. The perturbation degrades an exact-match mechanism's precondition, identical surface strings, in a way a fuzzy-matching mechanism is comparatively more robust to. We frame this as a property of this specific perturbation model and this specific retrieval implementation (rapidfuzz-based lexical similarity), not a claim about fuzzy or embedding-based retrieval in general, which this experiment does not test. This blindness is a direct consequence of how ADS is defined, so the qualitative shape of 6b is partly built into the construction. What that construction does not give is the magnitude and direction actually observed. The empirical winner is constant across all 120 conditions of each lexical condition (retrieval under VARIED, tie under CLEAN), and the rules-minus-retrieval gap widens, rather than narrows, as realized ADS rises.

6.5 Relationship to Algorithm Selection and Meta-Learning

This finding refines, rather than contradicts, the algorithm-selection and meta-learning lineage (Rice, 1976; Smith-Miles, 2009; Barbudo et al., 2023) by identifying a specific failure mode of a single-feature, label-consistency-only design-time selector: such a selector can be a good predictor of a mechanism's own expected difficulty while carrying no information about a separately governed ranking axis. Nothing here suggests that literature is wrong or superseded, only that this narrow selector class has a specific, empirically observed blind spot in this experiment.

6.6 Implications for Practice and Future Selector Design

Four inferences this experiment does not license, stated explicitly rather than left implicit. It does not follow that consistency-style signals are useless: the accuracy-prediction finding (Section 5.2) still holds. It does not follow that this result generalizes to other lexical-noise models or other retrieval implementations, neither of which was tested here. It does not follow that a fix exists: no decision rule conditioning on both consistency and a representation-stability signal was designed, prototyped, or tested anywhere in this repository (Section 8.1). And it does not follow that production data confirms this finding: production never ran the lexical-noise manipulation this experiment depends on (Section 1.1, Section 7.4).

The natural next step this experiment motivates is a decision rule that conditions mechanism selection on both realized consistency and a measured, not assumed, representation-stability signal. We name this only as a

direction, in future tense, with no pseudocode, no proposed threshold values, and no partial results, because none exists: presenting any implementation detail beyond "condition on both signals" would misrepresent an untested idea as a scoped method.

7 Limitations

7.1 H1 Only Partially Supported

H1 (revised) is only partially supported, matching the pre-registered PARTIALLY_SUPPORTED row of the falsification table exactly, stated here as the honest verdict, not softened toward either "confirmed" or "refuted." The mechanistic account in Section 6.4 is a well-evidenced, exhaustive, but post-hoc explanation of already-frozen data, not a fresh prospective confirmation. Retrieval's coverage is 1.0 in all 240 conditions, while rules' coverage is always below 1.0, a real asymmetry that is part of the explanation for the results reported in Section 5, not normalized away.

7.2 Synthetic Generator and Perturbation Model Scope

This experiment uses a single synthetic generator family; its 240 conditions replicate a seeded random-number generator, not an external population. The lexical perturbation used ($P_{\text{TRANSFORM}} = 0.3$, five fixed transform types) was pilot-tuned to a target corruption-share band, not derived from measured real-world OCR or typo error rates. The practical-equivalence margin ($\delta = 0.02$) that defines a tie versus a defined winner (Section 4.6) was likewise fixed once at calibration and never varied against the frozen outcome data; because this margin is the specific reason the CLEAN condition contributes zero defined comparisons (Section 5.6), this paper cannot rule out that a different margin would yield defined CLEAN-condition comparisons, a sensitivity this paper does not test.

7.3 Domain and Mechanism Scope

The motivating case study is drawn from one domain, Romanian fiscal-document invoice-to-account classification. The experiment itself is domain-agnostic by construction (Section 3.1), but has not been run against a second domain's data. Only exact-match rules and rapidfuzz-based retrieval were compared. Neither embedding-based retrieval nor the shipped multi-tier production cascade was tested. The large-language-model mechanism was

excluded from H1 for a documented, principled reason (the synthetic product string carries no signal predictive of the true label) and was not evaluated at all in Experiment 1.

7.4 Production Data Limitations

The production consistency figures cited in Section 1.1 (91.2% deterministic products, weighted ADS 0.847, unweighted ADS 0.964) were generated by an earlier version of the measurement pipeline and remain likely understated and unverified; they cannot be re-run against the corrected pipeline, since no production data exists in this repository. Production never ran a lexical-noise sweep; only two single-run data points feed the motivating "R3 flip" narrative in Section 1.1. Production data does not independently validate this experiment's finding, and this paper does not claim that it does.

7.5 No Selector Fix and Limited Generalization

No decision rule conditioning on both realized consistency and a representation-stability signal was designed, prototyped, or tested anywhere in this repository; Section 8.1 names it only as a future direction. The realized-ADS ceiling, a consequence of the fixed cross-company-alignment parameter, means the region at or above roughly 0.93 was never reachable or tested by this generator. That region is untested, not merely unfavorable. This paper makes no deployment or generalization claim: the finding describes this experiment, not a general recommendation for how to build classification systems.

8 Future Work

8.1 Representation-Stability-Aware Selectors

A decision rule that conditions mechanism selection on both realized ADS and a measured, not assumed, representation-stability signal is the direct next hypothesis this experiment motivates. Nothing of this kind was designed, prototyped, or partially tested here; it is named only.

8.2 Generalization: Additional Domains and Mechanisms

Re-running the corrected pipeline against a second domain's data would test whether the domain-agnostic design of Section 3.1 actually transfers, which this paper does not itself test.

Whether the specific pattern reported in Section 5.5 (retrieval's noise cost staying flat while rules' noise cost grows with realized ADS) is specific to rapidfuzz-style token similarity, or holds for other retrieval implementations such as embedding-based retrieval, is untested here.

8.3 Real-World Noise Models and Independent Datasets

A real, rather than synthetic, noise or OCR-error model calibrated against an actual document corpus would test whether the pattern observed under $P_{\text{TRANSFORM}} = 0.3$ generalizes beyond this pilot-tuned synthetic stand-in.

Re-running this experiment's design against an independently generated dataset, not merely a different seed of the same generator, would test whether the accuracy/ranking finding is a property of this generator family specifically.

8.4 Broader Architecture-Selection Experiments

A baseline comparison against simpler pipelines, and a before/after measurement of the production feedback loop, are both recommended in prior project documentation for upgrading currently weaker claims, but neither is required for this paper's narrower, locked contribution and neither is planned, scheduled, or begun.

9 Conclusion

This paper tested, with a pre-registered, 240-condition synthetic factorial experiment, whether historical decision consistency predicts which of two classification mechanisms, exact-match rules or fuzzy retrieval, should be selected under a controlled lexical perturbation. Realized historical consistency strongly predicts each mechanism's own accuracy. It does not predict which mechanism wins: the unconditional form of the original hypothesis is falsified, exceptionlessly, in the realized ≥ 0.90 band, and H1 overall is only partially supported, not confirmed. The main lesson is that historical consistency is informative about mechanism *difficulty*, not mechanism *ranking*, when, as observed here, ranking is governed by a representation-stability property the consistency signal does not observe. What remains open is whether a decision rule conditioning on both consistency and a measured representation-stability signal would perform better. We name it here as a direction, not as something built or tested. The practical consequence is narrow and concrete: a design-time selector reading only a consistency

statistic is most confident exactly where it is most wrong. In the realized ≥ 0.90 band, where R3's recommendation for rules is strongest, it disagreed with the empirical winner in all 18 tested conditions.

Reproducibility Statement

The generator, both mechanisms, the lexical perturbation model, and the calibration and analysis code used to produce every number in Section 5 are public and offline-reproducible with no client data, no API keys, and no cost; re-running `python scripts/experiments/exp1/run_final.py` reproduces `final_condition_results.csv` exactly. The production case study discussed in Sections 1.1 and 6.2 is confidential and not reproducible from this repository in any sense; only its cited aggregate statistics appear in this paper. That file is committed in the paper's public code repository, at `github.com/stiFFLer-codes/ADS-Cascade`, under `data/outputs/experiments/exp1/final/`. It is not bundled into the arXiv source package.

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